

Strong Coupling $\alpha_s(M_Z)$ and Two – Loop Renormalization Group Running

David A. Besemer^{1,*}

¹*Independent Researcher*

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We derive the strong coupling constant α_s at the Z – boson scale from the topological face diagonal of the 24 – cell lattice. We find the exact boundary condition $\alpha_s(M_Z) = \sqrt{2}/12 \approx 0.117851$. We implement a high – precision two – loop Runge – Kutta solver to verify the running of α_s up to the Planck scale, confirming asymptotic freedom.

I. INTRODUCTION

The strong coupling constant is the least precisely measured gauge coupling. We present a geometric derivation of its value at the Z mass.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\alpha_s(M_Z) = \frac{\sqrt{2}}{12} \approx 0.117851 \quad (1)$$

$$\mu \frac{d\alpha_s}{d\mu} = -\beta_0 \frac{\alpha_s^2}{2\pi} - \beta_1 \frac{\alpha_s^3}{8\pi^2} \quad (N_f = 5 \leftrightarrow 6 \text{ at } M_t) \quad (2)$$

III. TWO-LOOP RUNNING AND THRESHOLDS

We integrate the RGEs using a top-quark mass threshold $M_t = 173.1$ GeV. Below M_t , the number of active flavors is $N_f=5$, and above M_t , $N_f=6$.

IV. CONCLUSION

The two-loop running shows stable convergence to the Planck scale, validating the strong coupling's geometric boundary condition.

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- [3] D. A. Besemer, **sdgft** Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org